

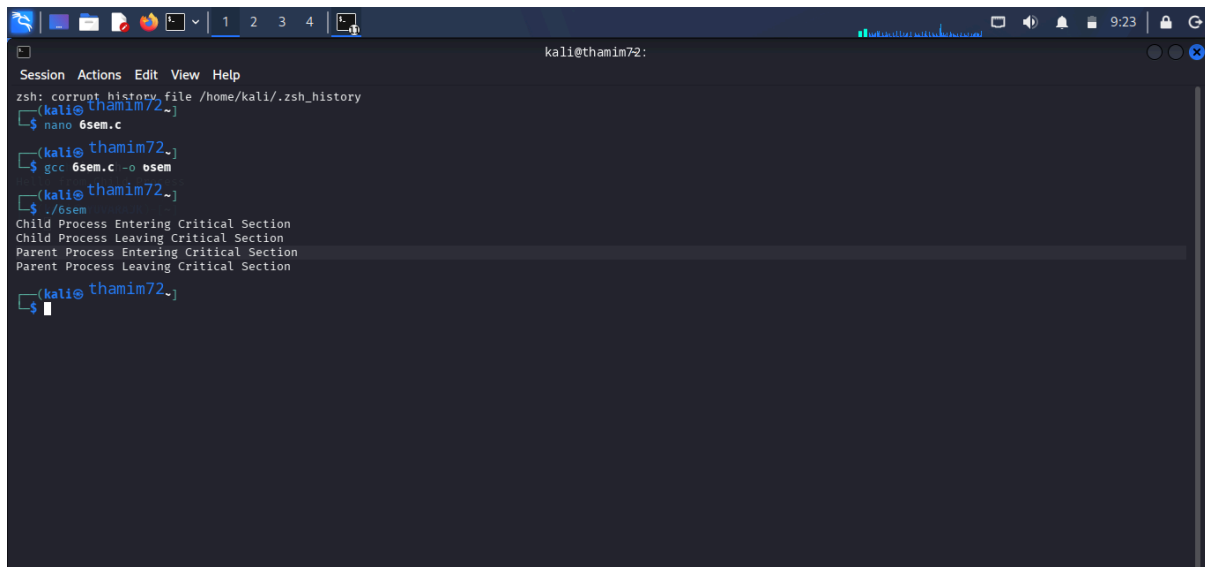
C PROGRAMMING :



```
GNU nano 9.0 6sem.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <semaphore.h>
#include <sys/mman.h>
#include <sys/wait.h>

int main()
{
    sem_t *sem;
    sem = mmap(NULL, sizeof(sem_t), PROT_READ | PROT_WRITE, MAP_SHARED | MAP_ANONYMOUS, -1, 0);
    sem_init(sem, 1, 1);
    if (fork() == 0) {
        sem_wait(sem);
        printf("Child Process Entering Critical Section\n");
        sleep(3);
        printf("Child Process Leaving Critical Section\n");
        sem_post(sem);
        exit(0);
    }
    sem_wait(sem);
    printf("Parent Process Entering Critical Section\n");
    sleep(3);
    printf("Parent Process Leaving Critical Section\n");
    sem_post(sem);
    wait(NULL);
    sem_destroy(sem);
    return 0;
}
```

OUTPUT :



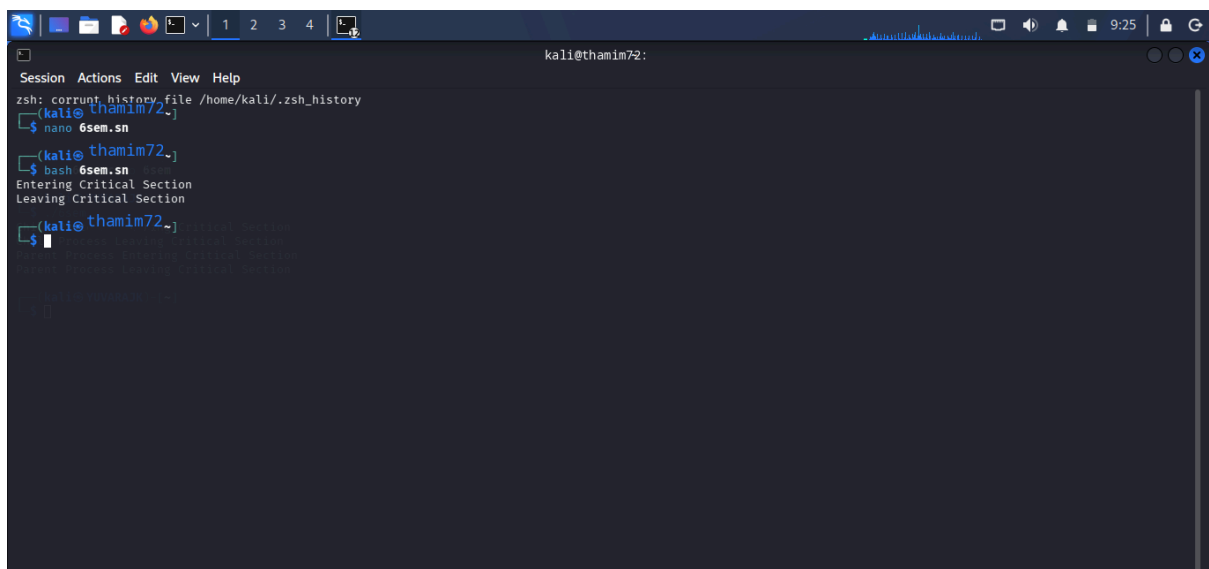
```
kali@thamim72:~$ nano 6sem.c
kali@thamim72:~$ gcc 6sem.c -o bsem
kali@thamim72:~$ ./bsem
Child Process Entering Critical Section
Child Process Leaving Critical Section
Parent Process Entering Critical Section
Parent Process Leaving Critical Section
kali@thamim72:~$
```

SHELL PROGRAMMING :



```
GNU nano 9.0 6sem.sh
#!/bin/bash
LOCKFILE="/tmp/mylock"
while [ -f "$LOCKFILE" ]
do
sleep 1
done
touch "$LOCKFILE"
echo "Entering Critical Section"
sleep 5
echo "Leaving Critical Section"
rm -f "$LOCKFILE"
sleep 1
done
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano 6sem.sn
(kali@thamim72~)
$ bash 6sem.sn
Entering Critical Section
Leaving Critical Section
(kali@thamim72~)
$
```